

Homework problem #3 MA(1) estimate θ ,

$$X_t = \theta_1 \varepsilon_{t-1} + \varepsilon_t$$

$$E[X_t X_{t-1}] = E[\theta_1 \varepsilon_{t-1} X_{t-1} + \varepsilon_t X_{t-1}]$$

$$= \theta_1 E[\varepsilon_{t-1} (\theta_1 \varepsilon_{t-2} + \varepsilon_{t-1})] + 0$$

$$= \theta_1 E(\varepsilon_{t-1}^2) = \theta_1 \sigma_\varepsilon^2$$

because ε_t, X_{t-1} uncorrelated

$$= \theta_1^2 \sigma_\varepsilon^2 + \sigma_\varepsilon^2 = (\theta_1^2 + 1) \sigma_\varepsilon^2$$

$$\rho(1) = \frac{\gamma(1)}{\gamma(0)} = \frac{\theta_1}{1 + \theta_1^2}$$

Solve for θ_1 in terms of $\rho(1)$
given $\rho(1)$ est θ_1

$$\frac{\sum x^2}{\sum x^2 x^2}$$

$$\begin{pmatrix} x_1 & \dots & x_n \end{pmatrix} \begin{pmatrix} x_1 & \dots & x_n \end{pmatrix} = \begin{pmatrix} x_1^2 & \dots & x_n^2 \end{pmatrix}$$

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